the other hand, the Commission acknowledges that while the number of potential self-aggregators might be small their overall trading volume might be large, because these market participants are also likely some of the highest trading-volume broker-dealers and registered investment advisors.

e. Competing Consolidators' Ability to Offer Differentiated Products

The last factor that may affect the reasonableness of the assumption that a sufficient number of competing consolidators will enter the market is the ability to offer differentiated products, determined by the demand for differentiated products and the feasibility of supplying differentiated products. The greater the ability to offer differentiated products, the more competing consolidators are likely to register until no economic incentives are left for new entry. In fact, the ability to differentiate may be necessary to ensure multiple competing consolidators can serve the market for the following reasons. As discussed above, the production of consolidated data involves relatively higher fixed costs (e.g., connectivity to the exchanges, data storage, technical infrastructure needed to process large amounts of data), and lower variable costs (e.g., costs of delivering the processed data to each customer).[2090] In such markets, the firms have additional incentives to increase the number of their customers in order to spread the fixed cost across a larger base of consumers. Without differentiation, the fixed cost nature of the market, and resulting economies of scale, could result in only one competing consolidator, because the largest competing consolidator would be able to offer the most competitive price.

---

would expect much of their "use case for direct feeds would be eliminated if the SEC's rule is implemented as proposed, and if there is a competitive consolidated tape offering with the processor physically located in the same data center as the broker/dealers" they employ as agents. See NBIM Letter at 4.

[2090] Some commenters agreed. See, e.g., Angel Letter at 3, 19; NBIM Letter at 2. See also supra Section V.B.3(a).

The Commission believes that differentiation will likely be possible both because market participants demand different market data products and services and because competing consolidators will have the incentives and ability to offer differentiated products to service those diverse needs.

Market participants' demand for consolidated market data products is heterogeneous because there are many different investor types (e.g., retail investors, small banks, market participants focused on value investment) that have differing investment strategies, and therefore, different data needs.[2091]

Additionally, competing consolidators will have the incentives and ability to differentiate their products to meet their customers' diverse needs. The Commission believes that competing consolidators will have strong incentives to offer differentiated products because of its potential implications for their survival in the market place.[2092] By offering products that are responsive to each type of customer's specific needs, competing consolidators can specialize and reduce their costs with this specialization. They can then pass these costs savings on to their customers as lower consolidation and dissemination fees and as a result capture market share. For example, competing consolidators could meet investors' diverse demand by offering different data products that range from the entirety of consolidated market data to subsets of consolidated market data such as top of book products.[2093] In addition, some competing consolidators could

---

2091 Several commenters agreed with the Commission. See, e.g., BestEx Research Letter at 5; NYSE Letter II at 9; IEX Letter at 9; MEMX Letter at 5.

2092 See infra Section V.C.2(b) for a discussion on the relationship between differentiation and prices.

2093 Several commenters agreed with the Commission that investors have diverse market data needs. See, e.g., IEX Letter at 9; MEMX Letter at 5; NYSE Letter II at 9; Angel Letter at 9.

differentiate themselves by specializing in lower latency data for a segment of the market where trading strategies require high speed data access. Other competing consolidators could target data users who might prefer not to have the lowest latency product if the higher latency products came with a lower price or additional analytics. Competing consolidators could offer a range of user interfaces and analytics (e.g., various ways to display consolidated data, or provide forecasting services) that appeal to different data users or could even offer an analytical environment to customize analytics (e.g., offer software tools allowing market participants to analyze and summarize consolidate data). Differentiation along these dimensions will allow competing consolidators to offer different services at potentially different prices to different types of end users.

Competing consolidators will also have the ability to differentiate because the amendments do not restrict the type or variety of products they can offer, which will be determined by competitive forces. Additionally, the amendments do not require competing consolidators to offer the entirety of consolidated data, potentially allowing them some fixed cost savings (e.g., on their connectivity and processing costs) if they offer narrower data content than the entirety of consolidated market data. However, there is some uncertainty about the extent to which competing consolidators can differentiate, because how fees are set by the effective national market system plan(s) might affect the feasibility to offer such diverse products.[2094] For example, while with differentiation competing consolidators can save on costs and lower their consolidation and dissemination fees, in the absence of differential prices for data content, competing consolidators' differentiated products will have smaller corresponding price

[2094] See infra Section V.C.2(a)(i)e for a discussion of the influence of fees on the ability to differentiate.

differences from their customers' perspective. This is because the biggest component of the overall data fees (i.e., data content, consolidation and dissemination, and connectivity fees) that the market participants pay will likely be data content fees that will go to the effective national market system plan(s). Thus, cost savings passed onto customers in terms of lower consolidation and dissemination fees will make a limited difference when customers are comparing overall data fees. As a result, potential competing consolidators will have a narrower price band within which to differentiate themselves and price their products.

Some commenters expressed concern that competing consolidators will not offer differentiated products.[2095] Some commenters said that market participants' ability to receive differentiated products depends on the choices of the Operating Committee(s) of the national market system plan(s) and competing consolidators, and that, the absence of differentiation will recreate the status quo.[2096] Another commenter stated that competing consolidators will not differentiate because this business will rely on economies of scale (i.e., achieving cost savings by increasing their scale), not on economies of scope (i.e., achieving cost savings by increasing their product offerings).[2097] The Commission believes that competing consolidators will offer differentiated products for two reasons.

First, while the Commission cannot be certain of whether such fees would provide discounts for those who wish to receive subsets of consolidated market data or based on usage

---

2095 See, e.g., NYSE Letter II at 4; Data Boiler Letter I at 79.

2096 See NYSE Letter II at 4; Nasdaq Letter IV at 25.

2097 See Data Boiler Letter I at 79.

categories,[2098] the Commission believes that some form of differential pricing for consolidated market data is the most likely outcome as discussed above.[2099] With differential pricing for the data content underlying consolidated market data, competing consolidators will have greater opportunity to offer differentiated products to market participants.[2100] Likewise, exchanges continuing to offer connectivity at different latencies with different corresponding prices would further promote product differentiation by competing consolidators. This is because differential connectivity fees will lead to different fixed costs for competing consolidators (e.g., competing consolidators specialized in serving higher latency customers can purchase slower or lower capacity connectivity products and lower their fixed costs), and thus, different consolidation and dissemination and connectivity fees can be charged to their customers. Finally, competing consolidators are not required to consolidate and disseminate the entirety of consolidated market data, for example if they want to concentrate on a customer segment that prefers narrower data content. All of these—differential data content fees, the exchanges' differential connectivity fees, and the lack of a requirement to process and provide the entire data content of consolidated market data—will allow a larger price band over which potential competing consolidators can differentiate and price their products to serve their customers' diverse needs. On the other hand, this differentiation can still take place, in a more limited way, even if the effective national market system plan(s) do not implement any differential data content fees.

2098 See infra Section V.C.2(b)(ii) for further discussion of these fees.

2099 See supra Section V.C.2(a)(i)b for a discussion on the potential new fee structures under the amendments.

2100 See supra Section V.C.2(b) for an additional discussion on differentiated products and data fees.

Second, the Commission believes that, for competing consolidators, scale and differentiation and specialization are complements not substitutes, as suggested by one of the commenters.[2101] Competing consolidators could expand their scale and market share to be able to spread their fixed costs over a larger set of customers than they otherwise would, by relying on their differentiated product offerings, similar to how the third party data aggregators operate today. For example, current third party data aggregators can be focused on more or less latency sensitive segments of the market and use this differentiation as a way to reach a larger set of customers than they otherwise would. The Commission believes that this business model will carry over into the new competing consolidator business, and could similarly differentiate across a variety of product characteristics such as latency, data content, analytics, and user interfaces.

Finally, in the absence of differentiation, the market might end up with only one competing consolidator;[2102] however, the Commission believes this is a low probability outcome for the reasons discussed above.

(ii) Risk of Few Competing Consolidator Registrants

As discussed in the previous section, there are several factors that may affect the number of competing consolidators entering the market. These factors determine the number of competing consolidators, which in turn determines the level of competition and ultimately the magnitude of benefits from the final amendments. While the Commission recognizes uncertainty in some of these factors, the Commission believes that it is reasonable to assume that there will be a

---

2101 See Data Boiler Letter I at 56.

2102 See infra Section V.C.2(a)(ii)a for a discussion on the probability and potential results of having a single competing consolidator operate in the market.

sufficient number of competing consolidators to achieve the benefits of the rulemaking and that the risk that the anticipated benefits of the amendments will not materialize because of insufficient competition among competing consolidators is low.[2103]

The assumption that there will be a sufficient number of competing consolidators entering the market affects some economic effects of the decentralized consolidation model. Generally, many of the benefits and competitive considerations below depend on this assumption. For example, the Commission believes that competition among competing consolidators will lead to lower fees paid by market participants for consolidated market data products,[2104] larger gains in efficiency in the delivery of consolidated market data products and market data communication innovations,[2105] as well as a reduction in data consolidation and dissemination latencies.[2106] In addition, some of the costs discussed below also depend on this assumption. For example, after the transition ends, the decentralized consolidation model will decrease regulatory compliance costs imposed by Regulation SCI on existing exclusive SIPs that may register as competing consolidators, by changing their systems from "critical SCI systems" to "SCI systems."[2107]

2103 One of the commenters did not "find any fault" with the Commission's assessment over the potential competitive outlook of the competing consolidator market. The commenter stated that "[t]he Department finds no fault with the SEC's preliminary determination that the risk is low that either no new SIP Data consolidators enter or only very few enter." See DOJ Letter at 5.

2104 See infra Section V.C.2(b).

2105 See infra Section V.C.2(c).

2106 Id.

2107 See infra Section V.C.2(e)(ii) for a discussion on the heightened requirements for "critical SCI systems" versus standard requirements for "SCI systems."

Some commenters questioned the Commission's assumption that there will be a sufficient number of competing consolidators and argued that there is not sufficient industry support for competing consolidators.[2108] On the other hand, several commenters indicated an interest in becoming a potential competing consolidator[2109] and one commenter predicted that several of the other current market participants will come forward to become one.[2110]

The Commission continues to believe that the risk that the anticipated benefits of the amendments will not materialize because the likelihood of insufficient competition among competing consolidators is low. Specifically, based on its analysis, as well as its experience and judgment, the Commission believes that there will initially be at least two competing consolidators and entry into the competing consolidator market space will likely continue until no economic incentives are left for any new competing consolidators to enter. The Commission believes that the most likely outcome is three or more competing consolidators with at least one competing consolidator that is not affiliated with either one of the exchanges currently operating the exclusive SIPs or an exchange that has sufficient proprietary data revenue that would create conflicting profit incentives.[2111] The Commission believes that this scenario will likely lead to vigorous competition and, as a result, will be enough for the predicted benefits to materialize.

a. Likelihood of Zero or One Competing Consolidator

2108 See, e.g., Nasdaq Letter IV at 2, 3, 23, 24, 47; NYSE Letter II at 13, 16; TechNet Letter II at 2; Angel Letter at 20; IDS Letter I at 3, 7. See also supra note 615.

2109 See, e.g., ACTIV Financial Letter at 1; McKay Letter at 2; NovaSparks Letter at 1; MIAX Letter at 1 for an expression of their interest in registering as competing consolidators.

2110 See NovaSparks Letter at 1.

2111 See infra Section V.C.2(a)(ii)a for a discussion on conflicting profit incentives of some potential competing consolidators.

In this section, the Commission analyzes the likelihood of zero or one competing consolidators registering and believes that the risk of either of these outcomes is low because of the strong incentives to enter. As such, and also because of the transition period requirements, the risk that the amendments will not achieve their benefits because only one or no competing consolidators register is low.

One commenter stated that the EU has been attempting to create a market for competing consolidators, but "no consolidators have signed up. By declaring that the risk of few or zero consolidators is low, the Commission appears to be signaling ignorance of the experience of other countries."[2112] The Commission does recognize the risk of no entry, but believes that strong incentives to enter render this risk low and that the European experience is not relevant to the U.S. because the regulatory framework in Europe is very different from that in the U.S.[2113]

2112 See Angel Letter at 20.

2113 The Commission does not believe that lack of potential consolidated tape providers (the EU equivalent of competing consolidators) in Europe has any implications for the U.S. markets or the predictions of the amendments because the regulatory framework within which the European market participants operate is very different from the U.S. Most significantly, the relevant European regulation, MiFID II, "does not mandate the establishment of a CT [consolidated tape] in the EU and does not oblige trading venues and APAs [approved public arrangements] to submit transaction data to a CTP [consolidated tape provider] for consolidation. The latter solution is the one chosen by the legislation in the US." Under the European regulatory framework, both the supply of and demand for market data would be uncertain, making it an economic calculation very different from the one the potential competing consolidators will make in the U.S. See European Securities and Markets Authority, *MiFID II/MiFIR Review Report No. 1*, at 35, available at https://www.esma.europa.eu/sites/default/files/library/mifid_ii_mifir_review_report_no_1_on_prices_for_market_data_and_the_equity_ct.pdf (last accessed Sept. 1, 2020). For additional discussion on how the European market data framework is different from the one in the U.S., see Philip Stafford, *EU-backed study calls for new body to track equities trades*, FINANCIAL TIMES (Oct. 7, 2020), available at https://www.ft.com/content/616acec6-cfc4-44d4-95c0-6053a041e0d7.

There is some risk of no entity entering the new competing consolidator business for two reasons and if no entity enters as a competing consolidator, none of the Commission's predicted benefits will materialize. First, the potential registrants with some of the lowest entry barriers are also the same market participants who expressed a strong preference to maintain the current status quo.[2114] Second, potential registrants who expressed interest in becoming competing consolidators also expressed some concern about not being able to compete with any potential competing consolidators affiliated with the exchanges operating the exclusive SIPs, because they might not be competing on a level playing field.[2115]

However, the Commission believes that this risk is low. Even if the two exchanges operating the exclusive SIPs choose not to become competing consolidators, there are several other potential entrants that stated that they are interested in becoming a competing consolidator.[2116] For example, one or more of the exchanges that do not currently operate an

---

2114 See, e.g., Equity Markets Association Letter at 2; NYSE Letter II at 24; Nasdaq Letter IV at 5. The two exchanges operating the exclusive SIPs stated that the exclusive SIP model performs very well and does not need to be replaced with the decentralized consolidation model.

2115 One commenter said that its letter discusses "the significance of establishing a level playing field by ensuring fair and equal access to exchanges and the need to extend these principles to the legs of the market data distribution system over which an exchange (or an exchange affiliate) may exercise direct or indirect control." See McKay Letter at 2. See also ACTIV Financial Letter at 2; MIAX Letter at 1.

2116 See, e.g., ACTIV Financial Letter at 1; McKay Letter at 2; NovaSparks Letter at 1; MIAX Letter at 1 for an expression of their interest in registering as competing consolidators. One market participant submitted a comment letter to an NYSE filing fee where the market participant stated that "Virtu plans to establish a competing consolidator to provide competitive market data products." See letter from Douglas A. Cifu, Chief Executive Officer, Virtu Financial, to Vanessa A. Countryman, Secretary, Commission, dated Aug. 28, 2020, available at https://www.sec.gov/comments/sr-nyse-2020-05/srnyse202005-7707480-222891.pdf. See also IEX Letter at 2, 3.